

Tag2Text: Guiding Vision-Language Model via Image Tagging

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<https://tag2text.github.io>

Abstract

This paper presents Tag2Text, a vision language pre-training (VLP) framework, which introduces image tagging into vision-language models to guide the learning of visual-linguistic features. In contrast to prior works which utilize object tags either manually labeled or automatically detected with a limited detector, our approach utilizes tags parsed from its paired text to learn an image tagger and meanwhile provides guidance to vision-language models. Given that, Tag2Text can utilize large-scale annotation-free image tags in accordance with image-text pairs, and provides more diverse tag categories beyond objects. As a result, Tag2Text achieves a superior image tag recognition ability by exploiting fine-grained text information. Moreover, by leveraging tagging guidance, Tag2Text effectively enhances the performance of vision-language models on both generation-based and alignment-based tasks. Across a wide range of downstream benchmarks, Tag2Text achieves state-of-the-art or competitive results with similar model sizes and data scales, demonstrating the efficacy of the proposed tagging guidance.

1. Introduction

Vision language pre-training (VLP) has shown an effective approach for learning a generic multi-modal representation and improving vision-language (V+L) tasks including generation-based (e.g., image captioning) and alignment-based (e.g., image-text retrieval). As large-scale datasets of image-text pairs [48, 4, 46, 41, 23] become available, recent works mainly focus on using transformer-based models to perform contrastive [41, 23, 30, 2, 29, 59] or generative learning [57, 29, 54, 6, 59, 55, 56] from massive image-text pairs. While great progress has been made, such studies normally rely on brute force pre-training manners that involve direct interaction with different modality features, modeling weakly-supervised learning due to the

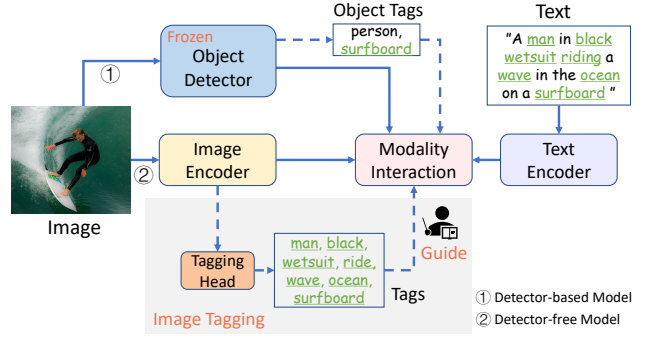


Figure 1. ①: Prior vision-language models primarily employ off-the-shelf object detectors to extract image features, restricting the capacity of the models. ②: Recent models generally utilize detector-free image encoders, ensuring a greater capacity. We introduce image tagging with a simple tagging head following the image encoder. The tagging head is supervised by annotation-free image tags parsed from its paired text. By integrating tagging guidance, our model achieves enhanced performance.

Method	No Manual Annotation	Enable End-to-End	Tag Categories	Additional Parameters	Running Time
Object Detector	✗	✗	Objects	$\geq 42M$	$\sim 153ms$
Image Tagging	✓	✓	Objects, Scenes, Attributes, Actions	$\leq 5M$	$\sim 40ms$

Table 1. Comparison of object detector and image tagging used in vision-language models.

lack of explicit alignment supervisions between image and text [32, 21, 60].

Prior approaches (e.g., OSCAR [32], VIVO [21], VinVL [61]) introduce the use of object tags as anchor points to ease the learning of semantic alignments between images and texts. However, these approaches rely on obsolete detector-based VLP frameworks, which employ off-the-shelf object detectors (e.g., Faster RCNN [42]) to extract image features (as shown in Figure 1 ①). The primary limitation of detector-based models is that the used object detectors are normally not perfect but have to be kept frozen during VLP to maintain detection ability, thereby restricting



BLIP:
A man riding a bike next to a train.
Tag2Text:
A man riding a bicycle next to a red passenger train on the tracks.
Tag2Text (User Specified):
A commuter train is passing under electric cables.



BLIP:
A group of people standing on top of a lush green field.
Tag2Text:
A group of young people playing a game of frisbee on a grassy field.
Tag2Text (User Specified):
The girl in yellow is looking at the frisbee player.



BLIP:
A painting of a turtle and a jellyfish.
Tag2Text:
A painting of a sea turtle swimming in the ocean with fish and jellyfish.
Tag2Text (User Specified):
A brown turtle and yellow fish under a blue sea watercolor.



BLIP:
A dog standing next to a cat on a couch.
Tag2Text:
A woman sitting on a couch with a brown dog standing next to her and an orange cat laying on the couch.
Tag2Text (User Specified):
A dog that is looking at the camera.

Figure 2. Comparison of image captioning results between Tag2Text (pre-training on 14M images) and BLIP [29] (pre-training on 129M images). Tag2Text integrates recognized image tags into text generation as guiding elements (highlighted in green underline), resulting in the generation with more comprehensive text descriptions. Moreover, Tag2Text permits users to input desired tags, providing the flexibility in composing corresponding texts based on the input tags.

the capacity of vision-language models [30, 11, 22]. The essential reason for this limitation is the inability of scaling up expensive grounding annotations. Moreover, utilizing object detectors leads to a substantial increase in model parameters and running time [30, 25].

As shown in Figure 1 ②, we propose a novel approach to provide tags for vision-language models by leveraging image tagging, and demonstrate that integrating image tagging with other VLP tasks in a multi-task manner is a natural and effective approach from two crucial perspectives. 1) **Data:** The pre-training image tags are obtained through automatically text semantic parsing, enabling large-scale annotation-free image tags in accordance with image-text pairs can be utilized, without the requirement of expensive grounding annotations which are necessary for object detectors. Image tagging also provides a better bridge between image and text, given that the parsed tag categories are more diverse and beyond objects, such as scenes, attributes, actions, etc. 2) **Architecture:** Image tagging merely necessitates an added recognition head followed by the original image encoder, ensuring efficient end-to-end pre-training and resulting in fewer parameters and improved efficiency. Table 1 provides a comprehensive comparison between image tagging and object detection.

Concretely, we present Tag2Text, a VLP framework which introduces image tagging into vision-language models to guide the learning of visual-linguistic features. Tag2Text achieves an exceptional tag recognition capability of 3,429 commonly human-used categories with only image-text pairs for training. More importantly, Tag2Text can effectively leverage tagging guidance to enhance the performance of vision-language models. **For generation-based tasks**, we design the training task as image-tag-text generation which empowers the model to produce text descriptions based on the image features in accordance with

assigned tags. As depicted in Figure 2, Tag2Text generates more comprehensive text descriptions with the guidance of comprehensively recognized tags. Additionally, Tag2Text permits users to input desired tags, providing the flexibility in composing corresponding texts [63]. **For alignment-based tasks**, while previous models rely on the alignment of multi-modality features which are considered black-box approaches, Tag2Text augments these methods by incorporating tags as visible alignment indicators.

On a wide range of downstream benchmarks, along with a series of qualitative results, Tag2Text demonstrates the efficacy of incorporating tagging guidance information into vision-language models. With similar model size and data scale, Tag2Text achieves state-of-the-art or competitive performance on multiple vision-language benchmarks including image captioning and image-text retrieval. Tag2Text also demonstrates superior performance on zero-shot multi-label recognition tasks, even comparable to supervised methods.

2. Related Work

Vision-Language Models consist of generation-based and alignment-based models. Generation-based models involve generating related text based on an input image. The initial approach of generation-based models relies on a two-stage process of recognizing tags from an image and then using them to compose a caption [13]. Notably, image features do not participate in the text generation stage. With remarkable progress of language models [9, 3, 39], language modeling gradually becomes a dominant pre-training objective in vision-language generation-based models [57, 29, 55, 6, 54, 56]. Such an approach endows vision-language models with the capability to generate expressive captions conditioned on visual information. Our proposed approach is

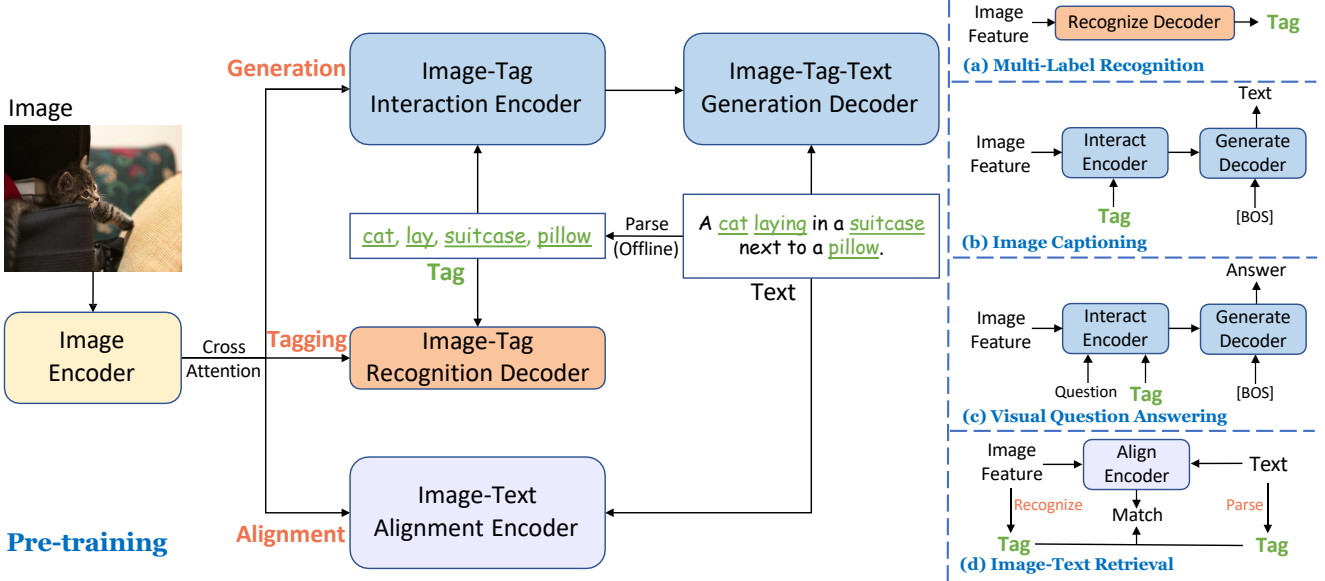


Figure 3. Illustration of Tag2Text framework. The core of Tag2Text lies in the introduction of image tagging supervised by the annotation-free image tags parsed from its paired text. **Generation:** Tag2Text learns to generate text related to the image by leveraging the automatically parsed tags, resulting in comprehensive and controllable texts with the guidance of recognized tags. **Alignment:** Tag2Text aligns the image and text, providing tags as visible alignment indicators during inference.

a novel scheme of image-tag-text generation, enabling our model to effectively regulate the content and quality of the generated text based on assigned tags.

Alignment-based models involve determining whether an image and a text are matched. Previous models perform either image-text contrastive learning [41, 23, 30, 2, 29, 22] with the dual-encoder architecture or image-text matching [32, 30, 2, 11, 29] with the fusion-encoder architecture. IDEA [22] introduces identified tags as additional text supervision only enhance image classification accuracy. These models predominantly rely on the alignment of multi-modality features which are considered black-box approaches for retrieval task. Tag2Text augments these methods by incorporating tags as visible alignment indicators, leading to further performance improvements.

Image Tagging, also known as multi-label image recognition, is a fundamental computer vision task that involves identifying multiple tags for a given image. Traditional approaches rely on a fully connected classifier and Binary Cross-Entropy loss (BCE) for optimization. Recent studies propose transformer-based classifiers [35, 45] to better leverage visual features, as well as robust loss functions [44, 62] to address the issues of missing samples and unbalanced positive-negative samples. Most existing multi-label datasets [34, 12] are based on manual annotations, which are labor-intensive and difficult to scale up. Our study employs text semantic parsing to efficiently obtain image tags and constructs a large-scale image tagging dataset comprising of 3,429 commonly used categories, re-

sulting in a superior tag recognition ability.

3. Approach

3.1. Overview Framework

We present Tag2Text, a VLP framework that enhances the performance of vision-language models by incorporating tagging guidance. Figure 3 shows the framework of Tag2Text. With large-scale image-text pairs, the core of Tag2Text lies in the utilization of image tags from texts. Initially, the image tags are extracted through text semantic parsing, providing a large-scale of tags without expensive manual annotations. Afterward, the parsed tags can serve as ground-truth labels for image tag recognition tasks. Moreover, we design a novel scheme of image-tag-text generation, enabling the model to effectively regulate the content and quality of the generated text with the guidance of recognized tags. Furthermore, Tag2Text encompasses image-text alignment and leverages tags as visible alignment indicators.

3.2. Mining Tags from Texts

Text Semantic Parser is adopted to parse text into image tags. The parser [58] first identifies *entities* ($= head + modifier$) and *relationships* from the input sentence based on the rules of the dependency tree. Subsequently, we obtain the tags (including *objects*, *scenes*, *attributes*, and *actions*) of the image based on the contrast maps from $head \rightarrow object/scene$, $modifier \rightarrow attribute$, and $relationship \rightarrow action$.

Tag Category System Construction is based on the principle that tags with higher frequency are considered more significant as they reflect common elements in the image descriptions. By employing the semantic parser, we process 4 million open-source image-text pairs and select the 5,000 most frequently occurring tags. Further filtering by human annotation results in the selection of the most commonly human-used 3,429 categories of tags. More statistics and details are presented in Section 4.1 and Appendix.

3.3. Tag2Text Pre-training

With triplet image-tag-text as inputs, Tag2Text employs a multi-task pre-training approach, which consists of Tagging, Generation, and Alignment. Both generation-based and alignment-based task utilize the guidance from image tagging to improve their performance. Concretely, the shared visual features obtained from the image encoder are interacted with various pre-training tasks through cross-attention.

Image Tagging aims to associate image features with the corresponding tags. As for the architecture, we apply an image-tag recognition decoder [35] to leverage visual spatial features, since tag-related visual concepts often distribute in different regions within an image. Due to the presence of missing labels in the parsed tags and an imbalanced distribution of positive and negative samples, we employ Asymmetric Loss (ASL) [44] for optimization.

Image-Tag-Text Generation aims to generate texts based on the image features in accordance with assigned tags. To achieve image-tag-text generation, Tag2Text employs the transformer encoder-decoder [50] architecture. The [BOS] token is prepended to the beginning of the text to indicate the start of a sequence. To eliminate positional bias, the image tags are rearranged prior to processing. Both tags and text are transformed into embeddings through tokenization and a word embedding matrix. The tag embeddings are integrated with image features in the image-tag interaction encoder and subsequently forwarded to the image-tag-text generation decoder for text generation. The pre-training objective is the unidirectional Language Modeling Loss (LM) to maximize the likelihood of the text in an autoregressive manner.

The distinction between image-tag-text generation and other generation approaches is illustrated in Figure 4. Our image-tag-text generation incorporate tags as a bridge to guide image features for text generation in an end-to-end manner. This approach enables the model to generate more comprehensive and controllable captions, provided that many accurate tags are given as the guidance signal.

Image-Text Alignment aims to determine whether a given pair of image and text is aligned. Following [29], Tag2Text leverages an additional image-text alignment encoder. The

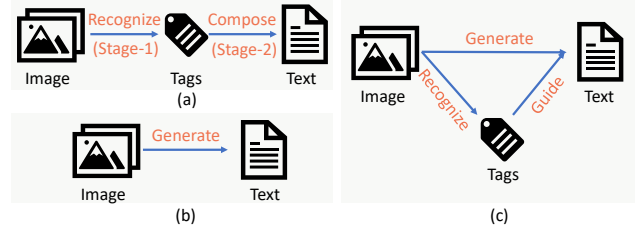


Figure 4. Image-text generation comparison. (a) Early works [13] primarily employ a multi-stage approach with separate tag recognition and text composition. The image features are not utilized during the text composition stage. (b) Recent works [57, 29, 6, 54] typically generate text directly from image features, which is challenging to control the generated text. (c) Our approach incorporates tags as a bridge to guide image features for text generation, improving content and quality control.

text is converted into embeddings via tokenization and word embedding. Then the text embeddings pass through the encoder and undergo coarse-grained Image-Text Contrastive Loss (ITC) with image features. Subsequently, the text embeddings undergo fine-grained Image-Text Matching Loss (ITM) with image features through cross-attention. The negative samples with higher ITC similarity will be selected for ITM with greater probability for hard mining.

3.4. Tag-Guided V+L Tasks

Multi-Label Recognition demands the model to recognize all relevant tags of an image. As illustrated in Figure 3(a), Tag2Text achieves this task by directly utilizing the image-tag recognition decoder.

Image Captioning entails the model to generate a textual description for a given image. Figure 3(b) shows that the same components for image-tag-text generation pre-training are utilized during finetuning. Previous image-text generation models are challenging to control the content of the generated description. By incorporating comprehensive tags recognized by the image-tag recognition decoder, our approach effectively improves the performance over the generated text. Furthermore, users can also input alternate guidance tags to generate descriptions highlighting specific aspects of the image.

Visual Question Answering aims to predict an answer to a question based on an image. Previous approaches typically treat VQA as a multi-class classification problem with a fixed set of answer choices. In contrast, Tag2Text employs an encoder-decoder architecture for generation, which is suited for generating free-form answers. As shown in Figure 3(c), the question, joint with tags, interacts with image features in the encoder, and then forwards to the decoder to generate a free-form answer.

Image-Text Retrieval encompasses both image-to-text and



Figure 5. Example results of Tag2Text on image-text retrieval. For each query text, the top 5 instances from the retrieval set are shown ranked from left to right. Tag2Text provides tags as additional visible alignment indicators (highlighted in green underline).

text-to-image retrieval. Previous methods match image-text pairs based solely on the features of different modalities, resulting in a lack of control and interpretability. Our approach, depicted in Figure 3(d), augments these methods by incorporating tags as visible alignment indicators ($Image \xrightarrow{Recognize} Tag, Text \xrightarrow{Parse} Tag$). By weighing the number of matching tags with feature similarity, Tag2Text boosts the retrieval results. Besides, real-world applications frequently involve users searching for images using several keywords rather than a sentence, highlighting the advantages of our approach.

4. Experiment

4.1. Experimental Setup

Pre-training Datasets. Following [30, 29], we pre-train our model on two widely used dataset settings, including a 4 million image dataset and a 14 million image dataset, respectively. The 4M image dataset setting includes two human-annotated datasets, COCO [34] and Visual Genome (VG [26]), and two web datasets, SBU Captions [38] and Conceptual Captions (CC-3M) [48]. The 14M image dataset setting builds upon the 4M setting by adding more noisy web dataset Conceptual 12M (CC-12M) [4]. The dataset statistics, including the number of texts and tags parsed from texts, are presented in Table 2.

Implementation Details. We adopt two most widely used backbones pre-trained on ImageNet [8] as the image encoder: ViT_{Base} [10] and Swin_{Base} [36]. Unless illustrated with subscript, the default model vision refers to ViT_{Base} as the image encoder. The encoder-decoder used for text generation and encoder for image-text alignment are 12-layer transformers [50] initialized from BERT_{Base} [9]. The generation decoder and alignment encoder share param-

	COCO	VG	SBU	CC-3M	CC-12M
#images	113K	100K	849K	2.81M	10.26M
#texts	567K	769K	849K	2.81M	10.26M
#avg texts	5.02	7.69	1.00	1.00	1.00
#tags	791K	607K	1.56M	5.93M	23.16M
#avg tags	7.00	6.07	1.84	2.11	2.26

Table 2. The statistics of the pre-training datasets.

eters with the cross-attention layers. The tag recognition decoder is a 2-layer transformer initialized from BERT_{Base} and shares parameters with the lowest 2-layer of the interaction encoder. The models are pre-trained for 20 epochs with the batch size of 960 on 8 NVIDIA A100 GPUs. The optimizer is the AdamW [37] with a weight decay of 0.05. The learning rate is warmed-up to $1e^{-4}$ in the first 3,000 iterations, and then follows linear decay with a rate of 0.9. The input images are resized to 224×224 uniformly during the pre-training stage.

4.2. Evaluation of Tagging Guidance

In Table 3, we verify the superiority of incorporating tagging guidance on a wide range of downstream benchmarks, including two generation benchmarks, two retrieval benchmarks, and two recognition benchmarks. Firstly, Tag2Text can be applied directly to multi-label recognition tasks with the additional tag recognition decoder. In Section 4.3, we demonstrate Tag2Text achieves superior tag recognition performance.

Based on this premise, the comprehensive tagging guidance provided by Tag2Text can effectively improve various vision-language (V+L) tasks. Tag2Text[†] w/o Tag Guidance refers to the commonly utilized text generation paradigm, which integrates image features into the text decoder through cross-attention layers. In text generation

Method	Pre-train #Images	Image Captioning				Image-Text Retrieval				Multi-Label Recognition	
		COCO-FT		NoCaps-ZS		COCO-FT		Flickr-ZS		COCO-FT	OpenImages-ZS
		B@4	CIDEr	B@4	CIDEr	TR@1	IR@1	TR@1	IR@1	mAP	mAP
<i>Generation-based model evaluation:</i>											
Tag2Text [†] w/o Tag Guidance	4M	35.1	117.5	23.0	85.3	✗	✗	✗	✗	✗	✗
Tag2Text [†]	4M	36.8 ^{+1.7}	121.6 ^{+4.1}	23.8 ^{+0.8}	87.2 ^{+1.9}	✗	✗	✗	✗	78.8	59.6
<i>Generation-based + Alignment-based model evaluation:</i>											
Tag2Text w/o Tag Guidance	4M	36.8	123.3	24.7	88.8	74.4	56.1	90.0	74.5	✗	✗
Tag2Text	4M	37.3 ^{+0.5}	124.6 ^{+1.3}	25.3 ^{+0.6}	89.5 ^{+0.7}	74.9 ^{+0.5}	56.6 ^{+0.5}	90.4 ^{+0.4}	75.2 ^{+0.7}	79.3	61.6
Tag2Text	14M	38.4	128.9	28.8	101.5	77.8	59.3	92.7	78.7	80.3	62.5

Table 3. Evaluation of tagging guidance on six downstream benchmarks including finetuning (FT) and zero-shot (ZS) settings. Tag2Text possesses tag recognition capability (multi-label recognition), and effectively integrates tag guidance to enhance the performance of vision-language models, including generation-based (image captioning) and alignment-based (image-text retrieval) benchmarks. In addition, Tag2Text enables data scale-up by leveraging training tags parsed from text without requiring manual annotation. Tag2Text[†] refers to no alignment tasks in the pre-training objectives. ✗ refers to the method which cannot be directly transferred to the corresponding benchmark.

Method	Pre-train #Images	Finetune on COCO	COCO-FT mAP	OpenImages-ZS mAP
<i>Training with full annotations of image tags:</i>				
Tagging	-	-	83.71	64.44
<i>Training with image-text pairs:</i>				
Tagging	4M	✗	75.43	53.34
Tag2Text	4M	✗	75.60	55.07
Tag2Text	✗	✓	76.54	36.16
Tag2Text	4M	✓	79.27	61.58
Tag2Text	14M	✓	80.25	62.46

Table 4. Comparison of multi-label recognition on COCO and OpenImages. Tag2Text achieves superior performance on zero-shot OpenImages (62.46 mAP), even comparable to the full supervision manner (64.44 mAP), which is trained with the complete annotations of OpenImages.

tasks, Tag2Text demonstrates a significant improvement across all metrics on both image captioning benchmarks compared to the approach which solely relies on image features. These findings are also evident in Figure 2, where the examples explicitly demonstrate the benefits of incorporating tag predictions. For instance, as depicted in the last example, the addition of identified tags, including “woman”, “sit”, and “orange”, results in the generation of a more comprehensive and coherent sentence.

Furthermore, our model adds image-text alignment objectives (ITC, ITM) for image-text retrieval tasks. Empirical results indicate that the alignment objectives boost the performance in text generation tasks as well. By integrating tags as visible alignment indicators into the foundation of image-text feature alignment, Tag2Text further enhances the results of two retrieval benchmarks. Figure 5 presents some examples of visible alignment indicators that enable effective alignment between image and text.

	Tag2Text	ML-Decoder (Multi-Label Recognition)	Detic (Object Detection)
	people, woman, man, frisbee, fence, disc, field, park, grass, game, play, stand, grassy, young, yellow	shoe, white, photograph, black, sneakers, T-shirt, human arm, person, adult, human hair, human, human head, social group, clothing, human face, human hand, male person, outerwear, mammal, human leg	person, jersey, jean, scarf, pole, frisbee, shoe, trousers, necklace, rubber band
	ocean, rock, water, bird, sand, beach, seagull, shore, flock, sky, fly, land, rocky, several, white, sandy	photograph, white, black, vertebrate body of water, sky	bird, person, boat, streetlight
	horse, house, road, carriage, street, people, ride, drive, pull, yellow, large, black, wagon, passenger, sit	photograph, white, vertebrate, equidae, home, mode of transport, property, carriage, wheel, mammal	horse, house, carriage, curtain, wheel, wagon, streetlight, person, awning

Figure 6. The comparison of recognized tags between Tag2Text and other state-of-the-art models for multi-label recognition (ML-Decoder [45]) and object detection (Detic [64]). Tag2Text offers more comprehensive and commonly used tags including objects, scenes, attributes, and actions (e.g., frisbee, fence, seagull, ocean, yellow, play, drive).

4.3. Analysis of Tagging Guidance

In this section, we present a detailed experimental analysis to investigate the effectiveness of tagging guidance.

Superior Tag Recognition Ability. To assess the tagging capability of Tag2Text, we conduct experiments on two multi-label recognition tasks: COCO [34] and OpenImages [28]. Our model is finetuned on the COCO training dataset and evaluated on the overlapping categories between Tag2Text and the two datasets, respectively. The tags used for finetuning are parsed from the text captions provided in the COCO Caption dataset, as the original Multi-Label COCO dataset only includes 80 categories of tags.

As demonstrated in Table 4, the integration of vision-language pre-training tasks into the model also improve the tag recognition ability. Furthermore, Table 4 highlights

Method	Pre-train #Images	COCO Caption Karpathy test		NoCaps validation							
		B@4	C	in-domain		near-domain		out-domain		overall	
				C	S	C	S	C	S	C	S
<i>Pre-trained with 4M images (COCO, VG, SBU, CC-3M):</i>											
DistillVLM [15]	4M	35.6	120.8	-	-	-	-	-	-	-	-
UFO [52]	4M	36.0	122.8	94.5	13.4	82.7	12.8	64.9	11.0	80.7	12.5
OSCAR [32]	4M	36.5	123.7	80.0	12.1	80.4	12.2	75.3	10.6	79.3	11.9
VIVO [21]	4M	-	-	90.4	13.0	84.9	12.5	83.0	10.7	85.3	12.2
BLIP* [29]	4M	37.0	123.6	91.3	13.5	86.4	13.0	84.2	11.8	86.6	12.8
ViTCap [14]	4M	36.3	125.2	98.7	13.3	92.3	13.3	95.4	12.7	93.8	13.0
Tag2Text-ViT	4M	37.3	124.6	95.1	13.5	89.0	13.0	86.9	12.1	89.5	12.9
Tag2Text-Swin	4M	38.4	128.7	101.0	13.9	96.0	13.6	96.7	12.9	96.9	13.5
<i>Pre-trained with more images:</i>											
Enc-Dec [4]	15M	-	110.9	92.6	12.5	88.3	12.1	94.5	11.9	90.2	12.1
MiniVLM [53]	11M	35.6	119.8	-	-	-	-	-	-	-	-
VinVL [61]	6M	38.2	129.3	103.7	13.7	95.6	13.4	83.8	11.9	94.3	13.1
LEMON [20]	12M	-	-	104.5	14.6	100.7	14.0	96.7	12.4	100.4	13.8
BLIP [29]	14M	38.0	127.8	-	-	-	-	-	-	102.2	13.9
Tag2Text-ViT	14M	38.4	128.9	104.8	14.6	100.4	13.9	102.5	13.3	101.5	14.0
Tag2Text-Swin	14M	39.1	131.8	106.7	14.5	105.7	14.4	110.7	14.3	106.9	14.4
BLIP+ <i>Bootstrap</i> [29]	14M	38.6	129.7	111.3	15.1	104.5	14.4	102.4	13.7	105.1	14.4
SIMVLM [57]	1.8B	39.0	134.8	-	-	-	-	-	-	94.8	13.1
LEMON [20]	200M	-	-	107.7	14.7	106.2	14.3	107.9	13.1	106.8	14.1

Table 5. Performance comparison of image captioning on the COCO and NoCaps Caption benchmarks. BLIP* refers to the result of our reproduction. +*Bootstrap* indicates the two stage dataset bootstrapping approach using generation and alignment tasks.

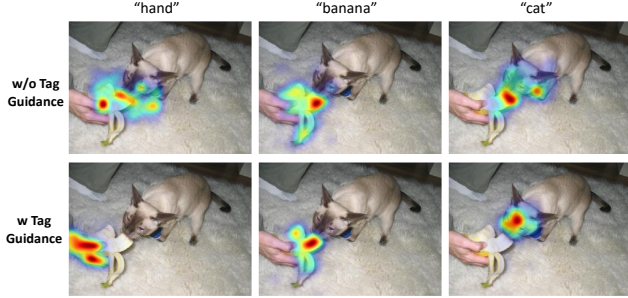


Figure 7. Grad-CAM visualization of image-tag interaction encoder. The tagging guidance can accurately locate corresponding image regions.

two-stage “pre-training + finetuning” paradigm in the context of multi-label recognition. The model, trained solely on the limited COCO dataset, fails to generalize well on the OpenImages dataset, attaining an mAP score of 36.16. However, when pre-trained on a large dataset, our model exhibits remarkable performance, even in the absence of any exposure to the training images from the OpenImages dataset, achieving an mAP score of 62.46, which is comparable to the fully supervised performance of 64.44 mAP.

Better Bridge between Image and Text. In order to highlight the superiority of Tag2Text in tag recognition, we compare the recognized tags with other state-of-the-art multi-label recognition and object detection open-source models. For multi-label recognition, we employ the ML-Decoder [45] model based on OpenImages [28] of 9,600 categories. For object detection, we employ the Detic [64] model based on LVIS [18] of 1,203 categories. The com-

parison results are illustrated in Figure 6, ML-Decoder recognizes many tags which are not frequently used and lacks many obvious common tags. On the other hand, Detic is limited to only recognizing object categories. In contrast, Tag2Text provides a more comprehensive and widely used set of tags, including objects, scenes, attributes, and actions (e.g., frisbee, fence, seagull, ocean, yellow, play, drive).

Accurate Image Region Localization. We expect the tags can effectively guide image features to generate corresponding text. To further verify this, we perform Gradient-Based Feature Localization (Grad-CAM) [47] on the image-tag interaction encoder. Results in Figure 7 verify that the tag embeddings accurately locate corresponding image regions, even for close image areas of tags such as “hand” and “banana”. This study reveals that Tag2Text achieves accurate image region location even without expensive grounding annotations.

5. Comparison with Other Methods

5.1. Evaluation on Image Captioning

The Image Captioning task encompasses two benchmarks: COCO Captions [24] and NoCaps [1], with the latter focusing more on recognizing novel objects. Both benchmarks are evaluated following the finetuning of the model on the training set of COCO Captions. The comparison of Tag2Text with other state-of-the-art generation models can be found in Table 5. To ensure fairness, we compare the results with base version of all methods without utilizing the CIDEr optimization [43].

Method	Pre-train #Images	COCO(5K test set)						Flickr30K(1K test set)					
		I2T			T2I			I2T			T2I		
		R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10
Pre-trained with 4M images (COCO, VG, SBU, CC-3M):													
UNITER [7]	4M	65.7	88.6	93.8	52.9	79.9	88.0	87.3	98.0	99.2	75.6	94.1	96.8
VILLA [16]	4M	-	-	-	-	-	-	87.9	97.5	98.8	76.3	94.2	96.8
OSCAR [32]	4M	70.0	91.1	95.5	54.0	80.8	88.5	-	-	-	-	-	-
METER-Swin[11]	4M	73.0	92.0	96.3	54.9	81.4	89.3	94.3	99.6	99.9	82.2	96.3	98.4
ALBEF[30]	4M	73.1	91.4	96.0	56.8	81.5	89.2	94.3	99.4	99.8	82.8	96.7	98.4
VLMO [2]	4M	74.8	93.1	96.9	57.2	82.6	89.8	92.3	99.4	99.9	79.3	95.7	97.8
BLIP* [29]	4M	75.0	92.7	96.2	56.9	81.9	88.9	95.0	99.6	99.9	81.9	96.0	98.0
OmniVL [51]	4M+Videos	76.8	93.6	97.3	58.5	82.6	89.5	94.9	99.6	99.9	83.4	97.0	98.6
Tag2Text-Vit	4M	74.9	92.5	96.2	56.6	81.5	88.8	94.3	98.9	99.6	80.5	95.5	97.6
Tag2Text-Swin	4M	77.5	94.1	97.2	60.0	83.3	89.9	94.8	99.6	100.0	84.2	96.7	98.5
Pre-trained with more images:													
FLAVA [49]	70M	61.5	82.1	89.6	50.1	74.4	83.2	85.4	95.7	98.3	73.2	92.7	95.5
UNIMO [31]	5.7M	-	-	-	-	-	-	89.4	98.9	99.8	78.0	94.2	97.1
METER-CLIP-ViT [11]	404M	76.2	93.2	96.8	57.1	82.7	90.1	94.3	99.6	99.9	82.2	96.3	98.4
ALIGN [23]	1.8B	77.0	93.5	96.9	59.9	83.3	89.8	95.3	99.8	100.0	84.9	97.4	98.6
ALBEF[30]	14M	77.6	94.3	97.2	60.7	84.3	90.5	95.9	99.8	100.0	85.6	97.5	98.9
BLIP [29]	14M	78.4	-	-	-	-	-	-	-	-	-	-	-
Tag2Text-Vit	14M	77.8	93.9	97.0	59.3	83.1	89.7	94.9	99.7	100.0	83.3	96.4	98.1
Tag2Text-Swin	14M	79.0	94.6	97.2	61.3	84.3	90.4	95.7	99.9	100.0	85.4	96.8	98.5
BLIP+Bootstrap [29]	14M	80.6	95.2	97.6	63.1	85.3	91.1	96.6	99.8	100.0	87.2	97.5	98.8

Table 6. Performance comparison of image-text retrieval on the COCO and Flickr30K benchmarks. BLIP* refers to the result of our reproduction. +Bootstrap indicates the two stage dataset bootstrapping approach using generation and alignment tasks.

Method	Pre-train #Images	test-dev	test-std
ViLT [25]	4M	71.26	-
FLAVA [49]	68M	72.80	-
UNITER [7]	4M	72.70	72.91
OSCAR [32]	4M	73.16	73.44
VILLA [16]	4M	73.59	73.67
UNIMO [31]	5.7M	75.06	75.27
ALBEF [30]	14M	75.84	76.04
Tag2Text-ViT	14M	75.32	75.41
Tag2Text-Swin	14M	75.84	75.82

Table 7. Performance comparison on the VQA v2 benchmark.

The experimental results demonstrate Tag2Text outperforms other methods across all metrics on both benchmarks with similar model size and data scale. Furthermore, Tag2Text surpasses most metrics of BLIP+Bootstrap, which employs a dataset bootstrapping approach, as well as LEMON and SIMVLM, which are pre-trained on 200 million and 1.8 billion images, respectively. Notably, due to the dual capability for both generation and alignment of Tag2Text, the performance of Tag2Text can also be further enhanced through Bootstrap, which we aim to accomplish in our future work.

5.2. Evaluation on Image-Text Retrieval

The Image-Text Retrieval task is evaluated on two benchmarks: COCO [34] and Flickr30K [40], for both image-to-text retrieval (I2T) and text-to-image retrieval (T2I). The performance comparison with other methods

are shown in Table 6. Under equivalent pre-training data and image encoder configurations, Tag2Text demonstrates comparable or superior performance compared to ALBEF, VLMO, and BLIP. Tag2Text-Swin leads to a further substantial improvement in performance. More importantly, the integration of tag alignment in Tag2Text makes it well-suited for practical search scenarios, where users search through the query of several keywords.

5.3. Evaluation on Visual Question Answering

We conduct experiments on the VQA v2 [17] benchmark for Visual Question Answering. Table 7 shows that Tag2Text outperforms or achieves competitive results with other approaches. During the experiment, we also find that incorporating tagging guidance does not improve the performance on VQA v2. We attribute this to the relatively simple nature of the questions, which typically have short answers. We anticipate more complex scenarios would highlight the value of tagging guidance and leave this exploration in future research.

6. Conclusion

This paper has presented Tag2Text, a vision-language pre-training framework, which introduces image tagging into vision-language models. Tag2Text achieves superior image tag recognition ability by exploiting fine-grained text information. Moreover, Tag2Text leverages tagging guid-

ance and effectively enhances the performance and controllability of vision-language models, including generation-based tasks and alignment-based tasks. On a wide range of vision-language tasks, Tag2Text demonstrates the value of tag as a bridge between image and text to infuse structure and knowledge information into vision-language models.

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A. Pre-training Objectives

Image Tagging. Image tagging is generally decomposed into multiple binary classification with binary cross-entropy loss (BCE) to optimize:

$$\begin{aligned}\mathcal{L}_{\text{Tagging}} &= -\mathbb{E}_{\mathbf{y} \sim D} [\text{BCE}(\mathbf{y}, P(\mathbf{y}))] \\ &= -\mathbb{E}_{\mathbf{y} \sim D} \left[\sum_{i=1}^C y_i \log P(\mathbf{y}_i) + (1 - y_i) \log(1 - P(\mathbf{y}_i)) \right]\end{aligned}\quad (1)$$

where y_i represents the label for the i -th category and C denotes the total number of categories. We employ Asymmetric Loss (ASL) [44] for optimization instead of BCE.

$$\begin{aligned}\mathcal{L}_{\text{Tagging}} &= -\mathbb{E}_{\mathbf{y} \sim D} [\text{ASL}(\mathbf{y}, P(\mathbf{y}))] \\ &= -\mathbb{E}_{\mathbf{y} \sim D} \left[\sum_{i=1}^C y_i (1 - P(\mathbf{y}_i))^{\gamma_+} \log P(\mathbf{y}_i) \right. \\ &\quad \left. + (1 - y_i) P(\mathbf{y}_i)^{\gamma_-} \log(1 - P(\mathbf{y}_i)) \right]\end{aligned}\quad (2)$$

Image-Tag-Text Generation. The pre-training objective for image-tag-text generation is Language Modeling Loss (LM) [3] to maximize the likelihood of the text in an autoregressive manner:

$$\begin{aligned}\mathcal{L}_{\text{LM}} &= -\mathbb{E}_{\mathbf{x} \sim D} [\text{CE}(\mathbf{x}, P(\mathbf{x}))] \\ &= -\mathbb{E}_{\mathbf{x} \sim D} \left[\sum_{i=1}^N \log P(\mathbf{x}_i | \mathbf{x}_{<i}) \right]\end{aligned}\quad (3)$$

where x_i represents the i -th token in the text and N denotes the total number of text tokens. Compared with the bidirectional Masked Language Modeling (MLM) [9], the unidirectional LM Loss is also gradually widely used in recent VLP studies [57, 54, 6, 29] as it enables seamless transfer of the model to text generation tasks.

Image-Text Alignment. The pre-training objectives for alignment utilizes Image-Text Contrastive Loss (ITC) based on multi-modal feature cos similarity [41, 30, 2, 29] and Image-Text Matching Loss (ITM) based on multi-modal feature fusion [30, 2, 29].

$$\mathcal{L}_{\text{ITC}} = -\mathbb{E}_{\mathbf{I}, \mathbf{T} \sim D} [\text{CE}(\mathbf{y}(\mathbf{I}), P(\mathbf{I})) + \text{CE}(\mathbf{y}(\mathbf{T}), P(\mathbf{T}))]\quad (4)$$

$$\mathcal{L}_{\text{ITM}} = -\mathbb{E}_{\mathbf{I}, \mathbf{T} \sim D} [\text{BCE}(\mathbf{y}, P(\mathbf{I}, \mathbf{T}))]\quad (5)$$

B. Tag Category Details

Tag Category System. We obtain annotation-free image tags parsed from its paired text. We construct our tag category system base on the principle that tags with a higher frequency are more significant, as they represent common elements in image descriptions. To this end, we process 4 million open-source image-text pairs (COCO, VG, SBU, CC-3M), and select the 5,000 most frequently occurring tags.

Further filtering by human annotation results in the selection of the most commonly human-used 3,429 categories of tags (including objects, scenes, attributes, actions). The tag category statistics are presented in Table 8. An illustration of the tag categories is provided in Figure 8, where the size of each word is proportional to the frequency of the category in the open-source image-text pairs.

	Object/Scene	Attribution	Action	Total
Categories	3,012	177	240	3429

Table 8. The statistics of tag categories recognized by Tag2Text.



Figure 8. Illustration of the most-frequent categories in our tag category system. The word size is proportional to the category frequency in the pre-training image-text pair dataset.

Compared with Public Datasets. This section provides the statistics of the overlap between our tag categories and other widely used public datasets. Table 9 shows the statistics for object/scene categories with other datasets including OpenImages [28], COCO [34], ImageNet [8], CIFAR100 [27]. Table 10 presents the statistics for action categories with other datasets including HICO [5], V-COCO [19], HOI-W [33]. To the best of our knowledge, we do not find appropriate public datasets for the recognition of attribute tag categories.

Categories	OpenImages	COCO	ImageNet	CIFAR100
Original	19982	91	1000	120
Overlapping	1988	73	358	94

Table 9. The statistics of object/scene categories overlapping with other public datasets.

C. Image Tagging Details

Downstream Task. We employ tags parsed from COCO Caption [34] for image tagging finetuning. Each image in



Figure 9. More image captioning results. Tag2Text integrates recognized image tags into text generation as guiding elements (highlighted in green underline), resulting in the generation with comprehensive text descriptions.

Categories	HICO	V-COCO	HOI-W
Original	117	23	9
Overlapping	90	26	9

Table 10. The statistics of action categories overlapping with other public datasets.

the COCO Caption dataset is accompanied by five descriptive sentences, offering a comprehensive description of the image. As a result, the tags parsed from these captions are considered to be a close approximation of a complete set of tag labels.

Due to many invalid download links, we collected 567 categories from the OpenImages [28] multi-label benchmark, consisting of 1,742,125 training images and 37,306 test images. Since Tag2Text can recognize 3,429 categories of tags, we respectively take the overlapping categories with COCO and OpenImages multi-label benchmarks for evaluation. There are 77 overlapping categories in 80 categories of the COCO benchmark and 301 overlapping categories in 567 categories of the OpenImage benchmark.

Tagging Head Comparison. Table 11 investigates the impact of various tagging recognition head on the performance of Tag2Text. The results show that transitioning from full connection to tag recognition decoder [35] results in improved performance in multi-label recognition, followed by

improvements in caption generation. This indicates that the enhancement of tagging recognition leads to improved text generation. To mitigate the increase in model parameters, we propose sharing the parameters of image-tag recognition decoder and image-tag interaction encoder, reducing the parameters and further boosting the performance.

Recognition Head	#Parameters	Caption-FT (COCO)	Caption-ZS (NoCaps)	Tagging-ZS (OpenImages)
Full Connection	392M	120.6	86.6	57.4
Recognition Decoder	409M	121.2	87.0	59.0
Recognition Decoder (Layer Shared)	394M	121.6	87.2	59.6

Table 11. Tagging recognition head comparison.

D. Image Captioning Details

More Example Results. In Figure 9, we show more examples of Tag2Text using tag guidance to generate comprehensive descriptions.

Finetuning Strategies Comparison. This section discusses two strategies employed by Tag2Text for image captioning finetuning based on input guidance tags, as illustrated in Figure 10 ① and ②. The first strategy, similar to image-tag-text generation in the pre-training stage, involves input guidance tags parsed from the paired text. This enables the model to utilize all available tags to create a comprehensive text description. However, Tag2Text usually rec-

ognize tags with similar meanings (*e.g.*, “man”, “person”), which may result in redundant sentences (*e.g.*, “a man ..., while a person ...”) with low evaluation metrics.

The second strategy involves using the same process as the inference stage, where the input guidance tags for finetuning are recognized by the model. This approach allows the model to select guidance tags for generating more precise text generation, as demonstrated in Table 12. In this paper, all evaluation metrics for image captioning of Tag2Text employ this strategy. In practical applications, since we observe users generally prefer more comprehensive descriptions, we utilize a mixed training approach that combines both strategies. In this paper, all visualization example results for image captioning of Tag2Text apply this mixed strategy approach.

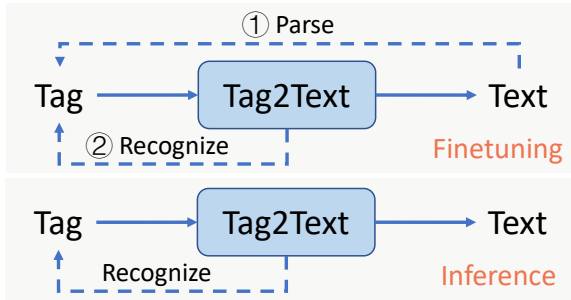


Figure 10. Illustration of two finetuning strategies for Image Captioning. By controlling the ratio of the two strategies, Tag2Text can generate more comprehensive (①) or more accurate (②) descriptions.

Input Guidance Tags	COCO		NoCaps-ZS	
	B@4	CIDEr	CIDEr	SPICE
Text Parsing	31.8	117.6	91.9	13.2
Model recognition	38.4	128.9	101.5	14.0

Table 12. Comparison of finetuning strategies for Image Captioning.